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| United States Patent | 12418718           |
| Kind Code            | B2                 |
| Date of Patent       | September 16, 2025 |
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### Image processing system and image processing method

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#### Abstract

An image processing system includes: an industrial machine; a visual sensor; a visual pattern arranged on a part to be detected, for identifying a relative positional relationship between the visual sensor and the visual pattern; an image processing unit that determines the relative positional relationship between the visual sensor and the visual pattern on the basis of an image of the visual pattern captured by the visual sensor; and a focusing operation control unit that, with the visual pattern within the field of view of the visual sensor, causes the visual sensor to capture the visual pattern while changing the relative positions of the visual sensor and the visual pattern in a predetermined direction with reference to the visual pattern, on the basis of the determined relative positional relationship to obtain a focus degree of the visual pattern, whereby the visual sensor is focused on the visual pattern.

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| <b>Appl. No.:</b>            | <b>18/043039</b>                                                 |
| <b>Filed (or PCT Filed):</b> | <b>December 03, 2021</b>                                         |
| <b>PCT No.:</b>              | <b>PCT/JP2021/044535</b>                                         |
| <b>PCT Pub. No.:</b>         | <b>WO2022/124232</b>                                             |
| <b>PCT Pub. Date:</b>        | <b>June 16, 2022</b>                                             |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20230328372 A1          | Oct. 12, 2023           |

## Foreign Application Priority Data

JP 2020-204868 Dec. 10, 2020

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## Publication Classification

**Int. Cl.:** H04N23/67 (20230101); G06T7/80 (20170101); H04N23/61 (20230101)

**U.S. Cl.:**

**CPC** H04N23/67 (20230101); G06T7/80 (20170101); H04N23/61 (20230101);

## Field of Classification Search

**CPC:** H04N (23/67); H04N (23/61); H04N (17/002); H04N (13/246); G06T (7/80); G06T (7/73); G06T (7/70); G06T (7/97); G05B (2219/39045); G05B (2219/39008); G05B (2219/39022); G05B (2219/39024); G05B (2219/37097); G05B (19/4086); G05B (2219/39029); G05B (2219/39016); G05B (2219/39047); G05B (2219/40003); G05B (2219/39527); G05B (2219/39397); B25J (9/1692); B25J (9/1697); G03B (43/00); G03B (13/36); Y10S (901/02); Y10S (901/14); Y10S (901/00); Y10S (901/01)

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## Background/Summary

### RELATED APPLICATIONS

(1) The present application is a National Phase of International Application No. PCT/JP2021/044535 filed Dec. 3, 2021, which claims priority to Japanese Application No. 2020-204868, filed Dec. 10, 2020.

### FIELD

(2) The present invention relates to an image processing system and an image processing method.

### BACKGROUND

(3) Systems configured to install a visual sensor on a movable part of an industrial robot or another industrial machine and perform visual inspection, position detection, and the like of a workpiece by image processing are known (for example, see PTL 1). In order to install a visual sensor on a movable part of an industrial robot or the like and perform position detection and the like of a workpiece, finding a relative positional relation of a coordinate system of the visual sensor with a standard coordinate system set to the industrial robot or the like, i.e., calibration of the visual sensor is required. Various techniques are known in the relevant field as technologies for calibration of a visual sensor (such as PTLs 2 to 5).

(4) In a robot system as described above, adjustment needs to be performed in order to focus a camera on an inspection target position. In such a case, the distance between a target object and the camera is generally adjusted by a user performing adjustment by turning a focusing ring of the camera or the user moving the robot by operating a teach pendant. On the other hand, PTL 5 describes that “an image processing system SYS determines a normal line V with respect to a set measurement point Wp on a determined workpiece W according to placement status of the workpiece W and varies the position and the attitude of a two-dimensional camera 310 in such a way that the determined normal line V matches the optical axis of the two-dimensional camera 310 (S1). By varying the distance between the two-dimensional camera 310 and the measurement point Wp in such a way that the determined normal line V matches the optical axis of the two-dimensional camera 310, the image processing system SYS focuses the two-dimensional camera 310 on the measurement point Wp.” (Abstract).

### CITATION LIST

#### Patent Literature

(5) [PTL 1] Japanese Unexamined Patent Publication (Kokai) No. 2018-91774 A [PTL 2] Japanese Unexamined Patent Publication (Kokai) No. 2014-128845 A [PTL 3] Japanese Unexamined Patent Publication (Kokai) No. H8-210816 A [PTL 4] Japanese Unexamined Patent Publication (Kokai) No. 2018-192569 A [PTL 5] Japanese Unexamined Patent Publication (Kokai) No. 2018-194542 A

### SUMMARY

#### Technical Problem

(6) While PTL 5 describes a configuration of performing focusing by adjusting the distance between a camera and a specific workpiece, the configuration in PTL 5 requires complex processing such as processing for specifying the shape of the workpiece, processing for determining the placement state of the workpiece, and processing for finding a normal line of the workpiece, and settings for the processing. Further, the workpiece may have various shapes, and therefore acquisition of placement information of the workpiece, acquisition of a normal line, and/or calculation of a degree of focusing may not be correctly performed.

#### Solution to Problem

(7) An aspect of the present disclosure relates to an image processing system including: an industrial machine; a visual sensor; a visual pattern for determining a relative positional relation between the visual sensor and the visual pattern placed at a detection target spot; an image processing unit configured to determine a relative positional relation between the visual sensor and the visual pattern from an image of the visual pattern captured by the visual sensor; and a focusing operation control unit configured to cause the visual sensor to focus on the visual pattern by causing the visual sensor to capture an image of the visual pattern while changing a relative position between the visual sensor and the visual pattern in a predetermined direction with the visual pattern as a basis, based on the determined relative positional relation, and finding a degree of focusing of the visual pattern, in a state of the visual pattern being captured in a visual field of the visual sensor.

(8) Another aspect of the present disclosure relates to an image processing method including: determining a relative positional relation between a visual sensor and a visual pattern placed at a detection target spot from an image of the visual pattern captured by the visual sensor; and causing the visual sensor to focus on the visual pattern by causing the visual sensor to capture an image of the visual pattern while changing a relative position between the visual sensor and the visual pattern in a predetermined direction with the visual pattern as a basis, based on the determined relative positional relation, and finding a degree of focusing of the visual pattern, in a state of the visual pattern being captured in a visual field of the visual sensor.

#### Advantageous Effects of Invention

(9) Placing a visual pattern at a detection target spot enables a visual sensor to automatically focus on the spot where the visual pattern is placed.

(10) The objects, the features, and the advantages, and other objects, features, and advantages will become more apparent from the detailed description of typical embodiments of the present invention illustrated in accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a device configuration of a robot system (image processing system) according to an embodiment.

(2) FIG. 2 is a diagram illustrating schematic hardware configurations of a visual sensor controller and a robot controller.

(3) FIG. 3 is a functional block diagram illustrating functional configurations of the visual sensor controller and the robot controller.

(4) FIG. 4 is a plan view of a calibration jig.

(5) FIG. 5 is a flowchart illustrating a focusing operation.

(6) FIG. 6 is a diagram illustrating an example of image data of an indicator captured by a camera.

### DESCRIPTION OF EMBODIMENTS

(7) Next, an embodiment of the present disclosure will be described with reference to drawings. In referenced drawings, similar components or functional parts are given similar signs. In order to facilitate understanding, the drawings use different scales as appropriate. Further, a configuration illustrated in a drawing is an example for implementing the present invention, and the present invention is not limited to the illustrated configuration.

(8) FIG. 1 is a diagram illustrating a device configuration of a robot system **100** according to an embodiment. As illustrated in FIG. 1, the robot system **100** includes an industrial robot (hereinafter referred to as a robot) **10** as an industrial machine, a robot controller **50** controlling the robot **10**, a visual sensor **70**, and a visual sensor controller **20** controlling the visual sensor **70**. The robot

system **100** is a system for recognizing the position of a workpiece, based on an image of the workpiece captured by the visual sensor **70**, and executing predetermined tasks such as inspection, handling, and machining of the workpiece. While the robot **10** is a vertical articulated robot in the example in FIG. **1**, another type of robot may be used.

(9) The visual sensor controller **20** and the robot controller **50** are connected through a communication interface and can exchange various types of information.

(10) A tool **31** as an end effector is installed on the tip surface of a flange **11** of an arm tip of the robot **10**. For example, the tool **31** is a hand for gripping a workpiece. The robot **10** executes a predetermined task such as handling of a workpiece in accordance with control by the robot controller **50**. The visual sensor **70** is installed on a supporting part of the tool **31** (i.e., a predetermined movable part of the robot **10**).

(11) The visual sensor **70** is controlled by the visual sensor controller **20** and captures an image of an image capture target such as a workpiece. A common two-dimensional camera may be used as the visual sensor **70**, or a three-dimensional sensor such as a stereo camera may be used. The visual sensor **70** includes an image sensor (such as a CCD or a CMOS) receiving a subject image and performing photoelectric conversion on the image, an optical lens condensing the subject image onto the image sensor, and the like. The visual sensor **70** transmits captured image data to the visual sensor controller **20**. Note that FIG. **1** illustrates an image M1 of a calibration jig J captured by the visual sensor **70** in a state illustrated in FIG. **1**.

(12) FIG. **2** is a diagram illustrating schematic hardware configurations of the visual sensor controller **20** and the robot controller **50**. The visual sensor controller **20** may be configured as a common computer in which a processor **21** is connected to a memory **22** (including a ROM, a RAM, a nonvolatile memory, and the like), an input-output interface **23** for communicating with an external device, and the like through a bus, as illustrated in FIG. **2**. The robot controller **50** may be configured as a common computer in which a processor **51** is connected to a memory **52** (including a ROM, a RAM, a nonvolatile memory, and the like), an input-output interface **53** for communicating with an external device, an operation unit **54** including various operation switches and the like, and the like through a bus. The robot controller **50** may be further connected to a teaching device (such as a teach pendant) for performing teaching of the robot **10**. While a device configuration in which the visual sensor controller **20** and the robot controller **50** are separate devices is illustrated in the present embodiment, the visual sensor controller **20** and the robot controller **50** may be configured to be the same device. For example, functions of the visual sensor controller **20** may be incorporated into the robot controller **50**.

(13) A robot coordinate system C1 is set to the robot **10**. The robot coordinate system C1 is a control coordinate system for controlling operation of each movable element in the robot **10** and is fixed in a three-dimensional space. While it is assumed as an example in the present embodiment that the origin of the robot coordinate system C1 is set at the center of a base part of the robot **10**, the robot coordinate system C1 may be set to have a different position and a different attitude.

(14) A flange coordinate system C2 is set on the tip surface of the flange **11**. The flange coordinate system C2 is a control coordinate system for controlling the position and the attitude of the flange **11** in the robot coordinate system C1. While it is assumed as an example in the present embodiment that the origin of the flange coordinate system C2 is placed at the center of the tip surface of the flange **11** and a z-axis of the system is set in such a way as to match the central axis of the flange **11**, the flange coordinate system C2 may be set to have a different position and a different attitude.

(15) When moving a wrist (tool **31**) of the robot **10**, the robot controller **50** (processor **51**) sets the flange coordinate system C2 in the robot coordinate system C1 and controls a servo motor of each joint in the robot **10** in such a way as to place the flange **11** at the position and the attitude indicated by the set flange coordinate system C2. Thus, the robot controller **50** can position the flange **11** (tool **31**) at any position and any attitude in the robot coordinate system C1.

(16) A sensor coordinate system C3 is set to the visual sensor **70**. The sensor coordinate system C3

is a coordinate system for defining coordinates of each pixel in image data captured by the visual sensor **70** and is set with respect to the visual sensor **70** in such a way that the origin of the system is placed at the center of a light-receiving surface (or an optical lens) of the image sensor in the visual sensor **70**, an x-axis and a y-axis of the system are placed parallel to a lateral direction and a longitudinal direction of the image sensor, and the z-axis of the system matches the line of sight (optical axis) of the visual sensor **70**.

(17) As will be described in detail later, the robot system **100** is configured as an image processing system performing a focusing operation by moving the visual sensor **70** in a predetermined direction with respect to a visual pattern for determining a relative positional relation between the visual sensor **70** and the visual pattern placed at a detection target spot of the visual sensor **70** by capturing an image of the visual pattern. According to the present embodiment, the detection target spot of the visual sensor **70** is the top surface of a base **60** on which a workpiece is equipped, and the visual pattern is formed on a calibration jig J. The visual pattern has only to have a visually recognizable form such as a pattern using paint or an engraved mark (unevenness). The visual pattern may be directly formed at the detection target position (the top surface of the base **60**). A visual pattern herein includes various visual displays such as a pattern, a mark, an indicator, an identification sign, and a symbol. Using the visual pattern in acquisition of a relative positional relation with the visual sensor **70** enables high-precision acquisition of a relative positional relation.

(18) In particular, precision can be enhanced, compared with a configuration of finding a relative positional relation between a camera and a specific workpiece as described in aforementioned PTL 5.

(19) The following description focuses on a function of the robot system **100** as such an image processing system.

(20) FIG. **3** is a functional block diagram illustrating functional configurations of the visual sensor controller **20** and the robot controller **50** when attention is focused on the function of the robot system **100** as an image processing system. As illustrated in FIG. **3**, the visual sensor controller **20** includes an image processing unit **202** executing image processing on an input image **201** captured by the visual sensor **70** and a calibration data storage unit **204** storing calibration data determining a relative position of the visual sensor **70** (sensor coordinate system C3) with respect to a standard coordinate system (the robot coordinate system C1 or the flange coordinate system C2) set to the robot **10**.

(21) For example, as a function of providing handling of a workpiece and the like, the image processing unit **202** has a function of detecting a workpiece from an input image by using a model pattern and detecting the position and the attitude of the workpiece. Furthermore, the image processing unit **202** according to the present embodiment includes a degree of focusing calculation unit **203** calculating a degree of focusing of a target object captured in a captured image.

(22) The calibration data storage unit **204** stores calibration data of the visual sensor **70**. For example, the calibration data storage unit **204** is a nonvolatile memory (such as an EEPROM). The calibration data of the visual sensor **70** include internal parameters and an external parameter of the visual sensor **70** (camera). The internal parameters include optical data such as lens distortion and a focal distance. The external parameter is the position of the visual sensor **70** with a predetermined reference position (such as the position of the flange **11**) as a basis.

(23) A geometric transformation characteristic inside the camera and a geometric relation between a three-dimensional space where an object exists and a two-dimensional image plane are found by using the calibration data, and furthermore, the three-dimensional spatial positions of the visual sensor **70** and the calibration jig J can be uniquely determined from a characteristic of the calibration jig J and an image of the calibration jig J captured by the visual sensor **70**. In other words, the position and the attitude of the calibration jig with the position of the visual sensor **70** as a basis can be found from information about the calibration jig J in an image captured by using the

calibrated visual sensor **70**.

(24) FIG. **4** is a plan view of the calibration jig **J**. Various calibration jigs known in the relevant field allowing the position and the attitude of a calibration jig to be found from an image captured by the visual sensor **70** with the position of the visual sensor **70** as a basis can be used as the calibration jig **J**. The calibration jig **J** in FIG. **4** is a jig allowing acquisition of information required for calibration of the visual sensor **70** by capturing an image of a dot pattern placed on a plane by the visual sensor **70** and satisfies the following three requirements: (1) grid point spacing of the dot pattern is known, (2) a certain number of grid points or more exist, and (3) a grid point to which each grid point corresponds is uniquely determinable. Without being limited to a jig in which a characteristic such as a predetermined dot pattern is placed on a two-dimensional plane as illustrated in FIG. **4**, the calibration jig **J** may be a jig in which a characteristic is placed on a three-dimensional solid body and has only to allow acquisition of three-dimensional positional information including positional information in a height direction (the direction of an arrow **91** in FIG. **1**) in addition to two-dimensional positional information (the X-direction and the Y-direction). Further, the calibration jig **J** may be the same jig that is used when the calibration data of the visual sensor **70** are acquired or may be different. Note that the internal parameters of the aforementioned calibration data are used in order to calculate the position and the attitude of a dot pattern with the position of the visual sensor **70** as a basis from an image of the dot pattern captured by the visual sensor **70**. While the calibration jig **J** including the dot pattern as illustrated in FIG. **4** is used as a target object for finding the position and the attitude of the target object viewed from an image capture device in the present embodiment, any type of target object allowing the position and the attitude of the target object viewed from the image capture device (visual sensor **70**) to be found may be used instead.

(25) The calibration jig **J** according to the present embodiment is installed at a target location on which the visual sensor **70** focuses (i.e., the top surface of the base **60**).

(26) As illustrated in FIG. **3**, the robot controller **50** includes an operation control unit **501** controlling operation of the robot **10** in accordance with an operation program. Furthermore, the robot controller **50** according to the present embodiment includes a focusing operation control unit **502** performing control of moving the robot **10**, based on a degree of focusing calculated by the degree of focusing calculation unit **203**, and moving the visual sensor **70** to an in-focus position.

(27) Note that a functional block in the visual sensor controller **20** or the robot controller **50** illustrated in FIG. **3** may be provided by executing various types of software stored in the memory by the processor (CPU) in the visual sensor controller **20** or the robot controller **50** or may be provided by a configuration mainly based on hardware such as an application specific integrated circuit (ASIC).

(28) FIG. **5** is a flowchart illustrating a flow of a focusing operation (image processing method) by the focusing operation control unit. It is assumed that calibration data of the visual sensor **70** are previously stored in the calibration data storage unit **204**. The focusing operation in FIG. **5** is executed under control by the processor **51** in the robot controller **50**.

(29) First, the focusing operation control unit **502** calculates and acquires a positional relation between the calibration jig **J** and the visual sensor **70**, based on an image of the calibration jig **J** captured by the visual sensor **70**. In this case, the focusing operation control unit **502** uses the calibration data stored in the calibration data storage unit **J**.

(30) By using the relative positional relation between the visual sensor **70** and the calibration jig **J** acquired in step **S1**, the focusing operation control unit **502** captures an image of the calibration jig **J** while moving the visual sensor **70** in such a way that the optical axis of the visual sensor **70** is parallel to the normal direction of the top surface of the calibration jig **J** (the direction of the arrow **91** in FIG. **1**) (step **S2**). At this time, the focusing operation control unit **502** causes the degree of focusing calculation unit **203** to calculate a degree of focusing of an image of a target object (i.e., dots of the calibration jig **J**) captured in the image (step **S3**).

(31) Then, by comparing the degree of focusing calculated by the degree of focusing calculation unit **203** with a reference value (degree of focusing reference value) as a value when the degree of focusing of the dots of the calibration jig **J** is highest, the focusing operation control unit **502** determines whether the degree of focusing acquired in step **S3** is the highest degree of focusing (step **S4**). The reference value (degree of focusing reference value) used here may be previously stored by the robot controller **50** (focusing operation control unit **502**); or for example, the focusing operation control unit **502** may perform an operation of moving the visual sensor **70** in a predetermined moving range and setting the highest degree of focusing as the degree of focusing reference value. When the degree of focusing is not determined to be highest (**S4**: NO), the focusing operation control unit **502** continues the processing of checking the degree of focusing while moving the visual sensor **70** (steps **S2** to **S4**). When the degree of focusing is determined to be highest (**S4**: YES), the focusing operation control unit **502** records the position of the visual sensor **70** when the degree of focusing is highest (in-focus position) as an image capture position (step **S5**).

(32) The degree of focusing calculation unit **203** may employ various calculation techniques known in the relevant field, such as a phase difference detection method and a contrast detection method, as the technique for calculating a degree of focusing. As an example, the degree of focusing calculation unit **203** may use a technique of, when reliability of a degree of focusing detected by a phase difference detection method is determined to be low, selecting a pixel range in which the phase difference detection method is used, based on a detection result by a contrast detection method, and by using a signal in the selected pixel range, calculating a degree of focusing by the phase difference detection method, as described in Japanese Unexamined Patent Publication (Kokai) No. 2013-29803 A. The degree of focusing calculation unit **203** may acquire data used for calculation of a degree of focusing from a functional component for calculation of focusing in the visual sensor **70**.

(33) A predetermined operating range in which the visual sensor **70** is moved in the processing in steps **S2** to **S4** may be determined based on an adjustment range of the focal distance of the camera. For example, the robot **10** may be controlled in such a way as to move the visual sensor **70** within a range of an image capture distance determined based on a focusing range of a focusing ring in the visual sensor **70**. Since the positional relation between the visual sensor **70** and the calibration jig **J** is known by step **S1**, such movement control of the visual sensor **70** is enabled.

(34) The image capture position recorded in step **S5** is used as an image capture position for positioning the visual sensor **70** when a predetermined task, such as visual inspection of a workpiece placed on the top surface of the base **60**, is executed. Thus, the focusing operation in FIG. 5 enables automatic positioning of the visual sensor **70** at a position where the sensor focuses on a position where the calibration jig **J** is installed. In other words, focusing of the visual sensor **70** can be automatically and efficiently performed.

(35) An example of calibration of the visual sensor **70** being previously performed (i.e., a case of the visual sensor controller **20** previously holding calibration data) has been described above; however, when calibration of the visual sensor **70** is not performed (i.e., in a case of the visual sensor controller **20** not holding calibration data), the robot controller **50** and the visual sensor controller **20** may acquire calibration data by using various calibration techniques (such as aforementioned PTLs 2 to 5) known in the relevant field.

(36) As described above, the calibration jig **J** is used in the focusing control operation in FIG. 5, according to the present embodiment, and therefore calibration can be easily performed on the spot by using the calibration jig **J** when the focusing control operation in FIG. 5 is performed. While a method for calibrating a camera is in itself a well-known technique and therefore description thereof is omitted, such a method is described in detail in, for example, "An Efficient and Accurate Camera Calibration Technique for 3D Machine Vision" by Roger Y. Tsai (CVPR, pp. 364 to 374, 1986 IEEE). As a specific means, a camera can be calibrated by causing a jig such as the



calibration jig J illustrated in FIG. 4 (a dot pattern plate on which dots are formed in a known geometric placement) to intersect with the optical axis of the camera at right angles and performing measurement at the positions of two known spots.

(37) An example of an image processing technique for determining the attitude of the sensor coordinate system C3 with a standard coordinate system set to the robot 10 (the robot coordinate system C1 or the flange coordinate system C2) as a basis will be described. The focusing operation control unit 502 may be configured to be further responsible for such a function. Making the attitude of the sensor coordinate system C3 with respect to the standard coordinate system set to the robot 10 known by the technique described here enables the optical axis of the visual sensor 70 to be directed in a predetermined direction (such as a normal direction) with respect to the calibration jig J in the aforementioned focusing operation.

(38) In this example, data of the attitude of the visual sensor 70 in the standard coordinate system are acquired based on image data of an indicator ID captured by the visual sensor 70. FIG. 6 illustrates an example of the indicator ID. The indicator ID is provided at a detection target position (the top surface of the base 60) and is composed of a circular line C and straight lines D and E orthogonal to each other. For example, the indicator ID has only to have a visually recognizable form such as a pattern using paint or an engraved mark (unevenness) formed on the top surface of the base 60.

(39) The robot controller 50 (focusing operation control unit 502) finds the attitude of the sensor coordinate system C3 with the standard coordinate system set to the robot 10 as a basis in accordance with the following procedures.

(40) (A1) An image of the indicator ID is captured and image data JD.sub.0 of the indicator ID are acquired in a state of the visual sensor 70 being placed at an initial position PS.sub.0 and an initial attitude OR.sub.0 by the robot controller 50 in such a way that the indicator ID enters the visual field of the visual sensor 70. It is assumed that an image (JD.sub.n) as illustrated in FIG. 6 is acquired.

(41) (A2) The image processing unit 202 acquires coordinates (x.sub.n, y.sub.n) of an intersection F from an image of the indicator ID captured in the image JD.sub.n as the position of the indicator ID and acquires the area of a circle C as a size IS.sub.n (unit: pixels) of the indicator ID. The image processing unit 202 acquires a size RS (unit: mm) of the indicator ID in a real space, a focal distance FD of the optical lens of the visual sensor 70, a size SS (unit: mm/pixel) of the image sensor as previously saved data.

(42) (A3) The image processing unit 202 acquires a vector (X.sub.n, Y.sub.n, Z.sub.n) by using the acquired coordinates (x.sub.n, y.sub.n), the acquired size IS.sub.n, the acquired size RS, the acquired focal distance FD, and the acquired size SS. Note that X.sub.n can be found from an equation  $X_{sub.n} = x_{sub.n} \times IS_{sub.n} \times SS / RS$ . Y.sub.n can be found from an equation  $Y_{sub.n} = y_{sub.n} \times IS_{sub.n} \times SS / RS$ . Z.sub.n can be found from an equation  $Z_{sub.n} = IS_{sub.n} \times SS \times FD / RS$ . The vector (X.sub.n, Y.sub.n, Z.sub.n) is a vector from the visual sensor 70 (i.e., the origin of the sensor coordinate system C3) to the indicator ID (specifically the intersection F) when the image data JD.sub.n are captured and are data indicating a relative position (or coordinates in the sensor coordinate system C3) of the indicator ID with respect to the visual sensor 70.

(43) (A4) Similarly, from an image JD.sub.1 of the indicator ID captured at a position PS.sub.1 acquired by translating the visual sensor 70 from the initial position by a predetermined distance  $\delta x$  in the x-axis direction of the flange coordinate system and at the attitude OR.sub.0, the image processing unit 202 acquires a vector from the visual sensor 70 to the indicator ID when the image data JD.sub.1 are captured.

(44) (A5) Similarly, from an image JD.sub.2 of the indicator ID captured at a position PS.sub.2 acquired by translating the visual sensor 70 from the initial position by a predetermined distance  $\delta y$  in the y-axis direction of the flange coordinate system and at the attitude OR.sub.0, the image

processing unit **202** acquires a vector from the visual sensor **70** to the indicator ID when the image data JD.sub.2 are captured.

(45) (A6) Similarly, from an image JD.sub.3 of the indicator ID captured at a position PS.sub.3 acquired by translating the visual sensor **70** from the initial position by a predetermined distance  $\delta z$  in the z-axis direction of the flange coordinate system and at the attitude OR.sub.0, the image processing unit **202** acquires a vector from the visual sensor **70** to the indicator ID when the image data JD.sub.3 are captured.

(46) (A7) From the data described above, the image processing unit **202** acquires a rotation matrix representing an attitude (W, P, R) of the visual sensor **70** (sensor coordinate system C3) in the flange coordinate system C2 by the following Equation (1).

[Math . 1]

$$(47) \quad M1 = \begin{pmatrix} (X_1 - X_0)/x & (X_2 - X_0)/y & (X_3 - X_0)/z \\ (Y_1 - Y_0)/x & (Y_2 - Y_0)/y & (Y_3 - Y_0)/z \\ (Z_1 - Z_0)/x & (Z_2 - Z_0)/y & (Z_3 - Z_0)/z \end{pmatrix} \quad (1)$$

(48) The robot controller **50** (focusing operation control unit **502**) may be further configured to find the position of the sensor coordinate system C3 with the standard coordinate system set to the robot **10** as a basis. Operation procedures in this case are described below.

(49) (B1) The robot controller **50** first sets a reference coordinate system C4 in the flange coordinate system C2 at the initial position PS.sub.0 and the initial attitude OR.sub.0. The robot controller **50** according to the present embodiment sets the reference coordinate system C4 in the flange coordinate system C2 in such a way that the origin of the reference coordinate system C4 is placed at the origin of the flange coordinate system C2 and the attitude (the direction of each axis) of the reference coordinate system C4 matches the attitude (W, P, R) acquired by the aforementioned procedure. Accordingly, the directions of the x-axis, the y-axis, and the z-axis of the reference coordinate system C4 are parallel to the x-axis, the y-axis, and the z-axis of the sensor coordinate system C3, respectively.

(50) (B2) Next, the robot controller **50** operates the robot **10** and places the visual sensor **70** (i.e., the flange **11**) at a position PS.sub.4 and an attitude OR.sub.1 by rotating the visual sensor **70** around the z-axis of the reference coordinate system C4 from the initial position PS.sub.0 and the initial attitude OR.sub.0 by an attitude variation  $\theta$ .sub.1 (first attitude variation).

(51) (B3) The image processing unit **202** operates the visual sensor **70** and captures an image of the indicator ID and acquires relative position data (X.sub.4, Y.sub.4, Z.sub.4) of the indicator ID with respect to the visual sensor **70** at this time, by a technique similar to that for finding the aforementioned attitude.

(52) (B4) Next, the robot controller **50** operates the robot **10** and places the visual sensor **70** at a position PS.sub.5 and an attitude OR.sub.2 by rotating the visual sensor **70** around the x-axis or the y-axis (i.e., an axis orthogonal to the direction of the line of sight) of the reference coordinate system C4 from the initial position PS.sub.0 and the initial attitude OR.sub.0 by an attitude variation  $\theta$ .sub.2 (first attitude variation).

(53) (B5) The image processing unit **202** operates the visual sensor **70** and captures an image of the indicator ID and acquires relative position data (X.sub.5, Y.sub.5, Z.sub.5) of the indicator ID with respect to the visual sensor **70** at this time, by a technique similar to that for finding the aforementioned attitude.

(54) Denoting a vector from the origin of the reference coordinate system C4 in the flange coordinate system C2 (the origin of the MIF coordinate system C2, according to the present embodiment) to the origin of the sensor coordinate system C3 the position of which is unknown by ( $\Delta X$ .sub.1,  $\Delta Y$ .sub.1,  $\Delta Z$ .sub.1), the following Equations (2) and (3) hold.

[Math. 2]

$$(55) \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \cdot \text{Math.} \begin{pmatrix} X_0 + X_1 \\ Y_0 + Y_1 \end{pmatrix} = \begin{pmatrix} X_4 + X_1 \\ Y_4 + Y_1 \end{pmatrix} \quad (2)$$

[Math. 3]

$$\cos \theta_2 \cdot \text{Math.} Y_0 - \sin \theta_2 \cdot \text{Math.} (Z_0 + Z_1) = Y_5 \quad (3)$$

(56) By solving the aforementioned Equations (2) and (3), the robot controller **50** can estimate the vector ( $\Delta X_{\text{sub.1}}$ ,  $\Delta Y_{\text{sub.1}}$ ,  $\Delta Z_{\text{sub.1}}$ ) from the origin of the reference coordinate system **C4** in the flange coordinate system **C2** to the origin of the unknown sensor coordinate system **C3**.

(57) As described above, the indicator **ID** is a visual pattern allowing determination of a relative positional relation between the visual sensor **70** and the indicator **ID**, and therefore the indicator **ID** may be used in place of the calibration jig **J** used in the aforementioned focusing operation.

(58) As described above, placing a visual pattern at a detection target spot can cause the visual sensor to automatically focus on a spot where the visual pattern is placed, according to the present embodiment. In other words, focusing of the visual sensor can be performed automatically and efficiently.

(59) While the present invention has been described above by using the typical embodiments, it may be understood by a person skilled in the art that changes, and various other changes, omissions, and additions can be made to the aforementioned embodiments without departing from the scope of the present invention.

(60) The configuration described in the aforementioned embodiment is applicable to a focusing operation in various industrial machines equipped with a visual sensor on a movable part.

(61) While the device configuration example illustrated in FIG. **1** is a configuration example in which the visual sensor is installed on the robot and the calibration jig is placed at a fixed position, a configuration in which the visual sensor is fixed in a workspace as a fixed camera and the calibration jig (visual pattern) is installed on the tool of the robot may be employed. A functional block configuration similar to the functional block configuration illustrated in FIG. **3** may also be applied in this case. In this case, the focusing operation control unit **502** causes the visual sensor to focus on the calibration jig (visual pattern) by causing the visual sensor to capture an image of the visual pattern while moving the visual pattern with respect to the visual sensor (while changing the relative position between the visual sensor and the visual pattern in a predetermined direction with the visual pattern as a basis), based on the determined relative positional relation, and finding a degree of focusing of the visual pattern, in a state of the visual pattern being captured in the visual field of the visual sensor.

(62) The placement of the functional blocks in the visual sensor controller **20** and the robot controller **50** in the functional block diagram illustrated in FIG. **2** is an example, and various modified examples may be employed as the placement of the functional blocks. For example, at least part of the functions as the image processing unit **202** may be placed on the robot controller **50** side.

(63) While an in-focus position is determined by comparing a degree of focusing with a degree of focusing reference value and determining whether the two match (step **S4**) in the focusing operation illustrated in FIG. **5**, an in-focus position may be acquired by moving the camera in a predetermined operating range and searching for a position where a peak of a degree of focusing is acquired, instead of such a configuration.

(64) A program for executing various types of processing including the focusing operation according to the aforementioned embodiment may be recorded in various computer-readable recording media (such as semiconductor memories including a ROM, an EEPROM, and a flash memory, a magnetic recording medium, and optical disks including a CD-ROM and a DVD-ROM).

REFERENCE SIGNS LIST

(65) **10** Robot **11** Flange **20** Visual sensor controller **21** Processor **22** Memory **23** Input-output interface **31** Tool **50** Robot controller **51** Processor **52** Memory **53** Input-output interface **54** Operation unit **60** Base **70** Visual sensor **100** Robot system **201** Input image **202** Image processing unit **203** Degree of focusing calculation unit **204** Calibration data storage unit

## Claims

1. An image processing system, comprising: an industrial machine; a visual sensor equipped on a predetermined movable part of the industrial machine; a visual pattern for determining a relative positional relation between the visual sensor and the visual pattern placed at a detection target spot of a workpiece; an image processing unit configured to determine the relative positional relation between the visual sensor and the visual pattern from an image of the visual pattern captured by the visual sensor; and a focusing operation control unit configured to cause the visual sensor to focus on the visual pattern by causing the visual sensor to capture the image of the visual pattern while moving, based on the determined relative positional relation, the visual sensor by the industrial machine to an in-focus position where the visual sensor focuses on the visual pattern to change a relative position between the visual sensor and the visual pattern in a predetermined direction with the visual pattern as a basis, and finding a degree of focusing of the visual pattern, in a state of the visual pattern being captured in a visual field of the visual sensor, wherein the visual sensor is calibrated in advance by using the visual pattern to obtain calibration data, and the focusing operation control unit is configured to use the calibration data representing a position and an attitude of the visual sensor with a predetermined reference position in the industrial machine as a basis to direct the visual sensor to the predetermined direction with the visual pattern as a basis.
  2. The image processing system according to claim 1, wherein the focusing operation control unit is configured to change the relative position between the visual sensor and the visual pattern in the predetermined direction with respect to a plane on which the visual pattern is formed.
  3. The image processing system according to claim 1, wherein the focusing operation control unit is configured to move the visual sensor in a predetermined operating range with respect to the visual pattern, and move the visual sensor to the in-focus position by comparing a degree of focusing reference value when a degree of focusing of the visual pattern is highest with the degree of focusing acquired while moving the visual sensor.
  4. The image processing system according to claim 1, wherein the industrial machine is an industrial robot.
  5. An image processing method, comprising: determining a relative positional relation between a visual sensor and a visual pattern placed at a detection target spot of a workpiece from an image of the visual pattern captured by the visual sensor, wherein the visual sensor is equipped on a predetermined movable part of an industrial machine; and causing the visual sensor to focus on the visual pattern by causing the visual sensor to capture the image of the visual pattern while moving, based on the determined relative positional relation, the visual sensor by the industrial machine to an in-focus position where the visual sensor focuses on the visual pattern to change a relative position between the visual sensor and the visual pattern in a predetermined direction with the visual pattern as a basis, and finding a degree of focusing of the visual pattern, in a state of the visual pattern being captured in a visual field of the visual sensor, wherein the visual sensor is calibrated in advance by using the visual pattern to obtain calibration data, and the method further comprises using the calibration data representing a position and an attitude of the visual sensor with a predetermined reference position in the industrial machine as a basis to direct the visual sensor to the predetermined direction with the visual pattern.
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